

Random Graphs

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Abstract

This dissertation explores the cross-section of graph theory and probability theory, a branch of mathematics called random graphs. This is concerned with graphs whose edges are determined probabilistically. This will primarily be explored using the binomial random graph model $G_{n,p}$, through which we will establish two fundamental results.

In order to approach this we must first introduce the necessary prerequisite knowledge, including key tools such as Markov's inequality and the second moment method. Once this is established we will prove the two main theorems. First, we prove Erdős's lower bound on the Ramsey numbers, $R(k) \geq 2^{\frac{k}{2}-1}$. Then we will prove that $\hat{p} = \frac{1}{n}$ is a threshold of the existence of a triangle in $G_{n,p}$, showing the probability of a triangle tends to 0 when p is much less than $\frac{1}{n}$ and that it tends to 1 when p is much greater than $\frac{1}{n}$.

Throughout the report, theoretical results are accompanied by computational experiments carried out through python. These will help build understanding of each development whilst verifying the theorems and illustrating threshold phenomena.

The primary reference for this work is the lecture notes on Random Graphs by Joshua Erde and supplemented by my own research.

Acknowledgements

I would like to thank my dissertation supervisor Alessandro Sisto for their support in understanding the content and guidance on its structure. I would also like to thank Alexandre Martin for stepping in as a supervisor when my own was not available. Andrew White my personal tutor has also helped me throughout this journey and was also the person who guided me towards topics like this through his course on biological modelling.

Use of GenAI

In the preparation of this dissertation I first encountered GenAI through spell checker that was already attached to overleaf. This was a hinderance as it was for some reason set to American English and so tried to convince me all my spelling was wrong. It was swiftly removed.

I also made use of the Generative AI tool Claude to verify some of my background information, debug code and identify potential references. This occurred when:

- I wanted to confirm that my two slightly different definitions of $\gg$ and $\ll$ actually meant the same thing.
- I questioned why my reference for my 1st listing was not working (I had added an extra]).
- I searched for where to find the original 1959 Erdős paper since he was referenced so often that the paper would not show up on the citation finders I was provided.

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1 Introduction

The idea of a random graph was first introduced in 1959 by Paul Erdős and Alfréd Rényi [8] during their investigations of the probabilistic method. Not only are random graphs useful in understanding graph behaviour, they are also applied more broadly to investigate complex combinatorial problems [12]. This then extends to real world applications in studying networks such as neural interfaces, the internet and social networks[13]. An example of a real world application is the modelling of disease spread through random graphs. Let each vertex represent an individual in a community, an edge between 2 people exists with probability p , representing the probability that they had sufficient contact to transmit the disease during a given period [14]. Although this model doesn't actually track the disease, the social network it models provides a framework for estimating if the disease will spread to a large portion of the population. It can also help determine if there is a critical threshold of contact probability $\hat{p}$ such that if the population has a sufficient rate of close contact, an epidemic becomes inevitable. This project will explore theorems that characterise structural properties of random graphs, such as threshold phenomena, which remain meaningful even as the scale of the graph grows large.

Under the term 'random graphs' there exist at least 3 models: the uniform random graph $G_{n,m}$, the binomial random graph $G_{n,p}$ and the Erdős-Rényi process $G_n(m)$ [9]. Throughout this paper, we will mainly focus on exploring the binomial random graph, although they are all equivalent under certain conditions.

The dissertation will use the following structure. We will begin by introducing the relevant background material from probability and graph theory. Then we will introduce random graphs, before exploring some key theorems and their proofs. These theorems will be accompanied by computational experiments that I have created to help illustrate and verify the content we discuss. The primary reference for this project is the lecture notes of Erdős [7], which has provided the foundation for the mathematics presented here.

2 Background Information

2.1 Graph Theory

We will begin by introducing some graph theory notation that will be used throughout this project.

Definition 2.1 (Graphs). A graph is a pair $X = (V, E)$, where $V = \{1, \dots, n\}$ denotes the set of *vertices*. E refers to the set of *edges*, and this set consists of pairs of vertices (V_1, V_2) .

We write $V(G)$ and $E(G)$ for the set of vertices and the set of edges and define $e(G) = |E(G)|$ and $v(G) = |V(G)|$.

Example 2.1 (Social Network, [13]). *Consider a classroom with n students. This can be modelled as a graph $G = (V, E)$ where each student is represented as a vertex in V and an edge connecting two students indicated a friendship between them. This model allows us to explore questions in a classroom network such as:*

- *Is there any isolated students?*
- *Who is the most popular student?*
- *Has the class divided into distinct social circles with no connections between them?*

Definition 2.2 (Clique). A set of vertices $S \subseteq V(G)$ is called a *clique* if every pair of vertices in S is connected by an edge. The *clique number* $\omega(G)$ is the size of the largest clique in G .

Example 2.2. *A graph with 3 vertices, all of which are connected is a clique. This is demonstrated in Figure 1.*

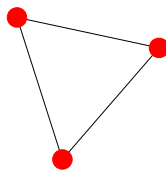


Figure 1: A Clique of size 3

Definition 2.3 (Independent Set). A set of vertices $S \subseteq V(G)$ is called a *independent* if no two vertices in S are connected by an edge. The *independence number* $\alpha(G)$ is the size of the largest independent set in G .

Example 2.3. *A graph with 3 vertices, none of which are connected by an edge is an independent set. This is shown in Figure 2*

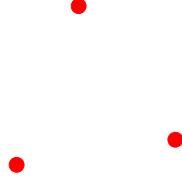


Figure 2: Independent set of size 3

2.2 Probability Theory

Definition 2.4 (Discrete Probability Space [10]). A discrete probability space is a pair $(\Omega, \mathbb{P})$ where:

- Ω is the countable set of all possible outcomes of an experiment, called the *discrete sample space*.
- $\mathbb{P}$ is a function $\mathbb{P} : \Omega \rightarrow [0, 1]$, called the *probability measure*.

A *random variable* X takes an outcome ω from the experiment and assigns to it a numerical value $X(\omega)$.

Definition 2.5 (Random Variable). The *random variable* X is a function mapping the sample space Ω to the set of real numbers $\mathbb{R}$:

$$X : \Omega \rightarrow \mathbb{R}.$$

It is known as a **Discrete Random Variable** if Ω is a discrete probability space. the image of the random variable is the countable set of possible values that the random variable can take.

Example 2.4. Let X be the exact height of a randomly selected person, then X is a random variable. This variable is not discrete since the height is not rounded, meaning it can take an infinite, uncountable number of possible values within a given range.

Example 2.5 (Discrete). Let Y be the number of heads attained from 3 independent coin tosses, then Y is a discrete random variable

The *expected value* is the weighted average of all possible events, weighted by its probability.

Definition 2.6 (Expectation). For a discrete random variable X :

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} p(\omega)X(\omega)$$

Definition 2.7 (Independence). Two events A and B are *independent* with respect to $\mathbb{P}$ if and only if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Example 2.6. Take 2 fair coins, let A be the event that the first coin lands on heads and B the event that the second lands on heads. Clearly $\mathbb{P}(A) = \frac{1}{2}$ and $\mathbb{P}(B) = \frac{1}{2}$.

$$\mathbb{P}(A \cap B) = \frac{1}{4} = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \mathbb{P}(A)\mathbb{P}(B),$$

so A and B are independent.

Example 2.7. Take a standard deck of cards, let A be the event that the first draw is a heart and B be the event that the second is also a heart. Then $\mathbb{P}(A) = \frac{1}{4}$, whilst $\mathbb{P}(B|A) = \frac{12}{51}$. Therefore,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B|A) = \left(\frac{1}{4}\right)\left(\frac{12}{51}\right) = \frac{1}{17}.$$

whereas

$$\mathbb{P}(A)\mathbb{P}(B) = \left(\frac{13}{52}\right)\left(\frac{13}{52}\right) = \frac{169}{2702}.$$

Since the two are not equal, the events are not independent.

Properties (Expectation). For any two random variables X and Y

$$1. \quad \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

Now let X and Y be independent, then

$$2. \quad \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

The most important random variable we will use in this paper is the *Binomial random variable* $\text{Bin}(n, p)$. It is formed from the sum of n mutually independent Bernoulli random variables $\text{Ber}(p)$, which take the value 1 with probability p and 0 with probability $(1 - p)$. Its expectation is np . If we assign $X = \text{Bin}(n, p)$ then we know that

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

Theorem 2.1 (Boole's Inequality/Union Bound, [2]). Let $E_1, E_2, \dots, E_n$ be a collection of events. Then,

$$\mathbb{P}\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n \mathbb{P}(E_i).$$

Also, if the collection of events are countably infinite then,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} \mathbb{P}(E_i).$$

Theorem 2.2 (Markov's Inequality, [11]). Let $X > 0$ be a random variable. Then for all k ,

$$\mathbb{P}(X \geq k) \leq \frac{\mathbb{E}(X)}{k}.$$

Theorem 2.3 (Cauchy-Schwarz Inequality [4]). *For any two random variables X and Y , we have*

$$|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}.$$

An important application of the Cauchy-Schwarz inequality is the *second moment method*. This method is a common technique used in probabilistic combinatorics showing that a non-negative random variable is positive through comparisons with its first and second moments.

Second Moment Method:[4]

Let X be a non-negative random variable, then

$$\mathbb{P}(X > 0) \geq \frac{(\mathbb{E}[X])^2}{\mathbb{E}[X^2]}.$$

Proof. If we start from the Cauchy-Schwarz inequality in the form

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2], \quad (1)$$

for random variables X, Y . Then if we let $Y = 1_{\{X>0\}}$ where $1_{\{X>0\}}$ is the indicator variable defined by

$$1_{\{X>0\}} = \begin{cases} 1 & \text{if } X > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Now we can sub this into each term in (1) as follows. First,

$$\mathbb{E}[XY] = \mathbb{E}[X1_{\{X>0\}}] = \mathbb{E}[X],$$

since $X \geq 0$. Now, since $1_{\{X>0\}}^2 = 1_{\{X>0\}}$ we have

$$\mathbb{E}[Y^2] = \mathbb{P}(X > 0),$$

by the fact that the expectation of an indicator function equals the probability of the corresponding event, see [4, Chapter 3]. Putting this together, inequality (1) becomes

$$(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2]\mathbb{P}(X > 0).$$

Rearranging this gives

$$\mathbb{P}(X > 0) \geq \frac{(\mathbb{E}[X])^2}{\mathbb{E}[X^2]}.$$

□

2.3 Useful Estimates

In Chapter 4 we will introduce the probabilistic method, in order to form proofs using this method we will often have to show examples that probabilities lie in the range $[0,1]$. This will require approximations to reduce difficult combinatorial expressions into something more manageable.

Big O notation will be used throughout this project. Formally, it is defined as $f(x) = O(g(x))$, for two functions $f(x)$ and $g(x)$, if and only if there exists constants C and N such that

$$|f(x)| \leq C|g(x)| \quad \forall x > N.$$

This means that $f(x)$ cannot grow faster than $g(x)$. See [1] for a more detailed explanation.

Example 2.8. Consider $f(x) = 3x^3 + x^2 + 5x + 1$ then

$$f(x) = O(x^3).$$

We also write $f(n) \ll g(n)$ if for every constant $C > 0$, we have $f(n) \leq Cg(n)$ for all sufficiently large n . Which means that $\frac{f(n)}{g(n)} \rightarrow 0$ as $n \rightarrow \infty$. Similarly, we write $f(n) \gg g(n)$ if for every constant $C > 0$ we have $f(n) \geq Cg(n)$ for all sufficiently large n . This means that $\frac{f(n)}{g(n)} \rightarrow \infty$ as $n \rightarrow \infty$.

For many of the proofs that will follow, the binomial coefficient $\binom{n}{k}$ will be present, since this is equal to $\frac{n!}{k!(n-k)!}$, exact values can be difficult to manage for large n and k . So we will now introduce some approximations to simplify expressions and to allow us to easily analyse how probabilities and expected values scale as $n \rightarrow \infty$.

A very weak upper bound of the coefficient is $\binom{n}{k} \leq n^k$ since n^k is the number of ways to pick k elements with repetitions. A much stronger estimate that will prove very useful is

$$\binom{n}{k} = \frac{n^k}{k!} + O(n^{k-1}).$$

See [6] for its derivation.

3 Random Graphs

The previous sections have introduced the relevant background material we will need for this project. We can now combine these to introduce the main focus of the dissertation, the random graph. Rather than focusing on analysing a given graph, we will consider a graph with individual edges determined by given probabilities. This enables us to study the likelihood that a randomly chosen graph has a specific property.

The simplest model of a random graph is the *uniform random graph*. Each uniform graph $G \in \Omega$ is assigned an equal probability.

$$\mu(G) = \frac{1}{|\Omega|} = \frac{1}{2^{\binom{n}{2}}}$$

since each of the $\binom{n}{2}$ possible edges can either be present or absent, which gives $2^{\binom{n}{2}}$ possible graphs.

3.1 Binomial Random Graph

The binomial random graph denoted $G_{n,p}$ is a random graph on n vertices where each of the $\binom{n}{2}$ possible edges appears independently with probability $p \in [0, 1]$.

Definition 3.1 (Binomial Random Graph, [5]). Let $V = \{1, 2, \dots, n\}$ be a set of vertices, and let Ω be the set of all graphs on V . For a given $p \in [0, 1]$, define a probability space on Ω by assigning to each graph $G \in \Omega$ the probability

$$\mathbb{P}(G) = p^{e(G)}(1-p)^{\binom{n}{2}-e(G)}.$$

This probability space is called the binomial random graph, denoted $G_{n,p}$.

The expected number of edges for this graph is:

$$E(X) = p \cdot \binom{n}{2}$$

Proposition 3.1. *The uniform random graph on vertex set $V = \{1, \dots, n\}$ is equivalent to $G_{n, \frac{1}{2}}$.*

Proof. The binomial random graph $G_{n, \frac{1}{2}}$ is equivalent to assigning an independent Bernoulli random variable $X_e \sim \text{Ber}(\frac{1}{2})$ to each pair $e \subseteq \{1, \dots, n\}$. Here $X_e = 1$ if an edge is present and equals 0 otherwise. Each X_e is pairwise independent, meaning for any two pairs e_1, e_2 and for all τ_1, τ_2 :

$$\mathbb{P}(X_{e_1} = \tau_1 \ \& \ X_{e_2} = \tau_2) = \mathbb{P}(X_{e_1} = \tau_1) \cdot \mathbb{P}(X_{e_2} = \tau_2)$$

. Therefore,

$$\mathbb{P}(X = G) = \mathbb{P}(X_e = 1 \ \forall e \in E(G) \ \& \ X_e = 0 \ \forall e \notin E(G))$$

becomes

$$\mathbb{P}(X = G) = \prod_{e \in E(G)} \mathbb{P}(X_e = 1) \prod_{e \notin E(G)} \mathbb{P}(X_e = 0) = \left(\frac{1}{2}\right)^{e(G)} \cdot \left(\frac{1}{2}\right)^{\binom{n}{2} - e(G)}.$$

Thus,

$$P(X = G) = \frac{1}{2^{\binom{n}{2}}}.$$

□

In order to simulate a random graph in Python I opted to model the graph through 1 matrix. In practise, this means creating a matrix with randomised binary entries. Take any element A_{ij} from this matrix, (i, j) is the pair of vertices and they have an edge if $A_{ij} = 1$. In order for this to accurately reflect a random graph, I set every entry other than the upper triangle to 0, before adding the new matrix to its transpose to formulate a symmetric matrix with diagonal entries of 0. Thus, simulating a random graph. The code for this can be seen below:

Listing 1: Random Graph Generator

```
import numpy as np

def X(n, p):
    upper = np.random.binomial(1, p, size=(n, n))
    #generate a nxn matrix, entries 0/1
    A = np.triu(upper, 1)
    #restricts to the upper triangle above the diagonal
    A = A + A.T #the matrix is now symmetric
    return A
```

Then I could use Python's Networkx package to help display the random graphs I had simulated using the algorithm found in the appendix. Below is an example of this simulation.

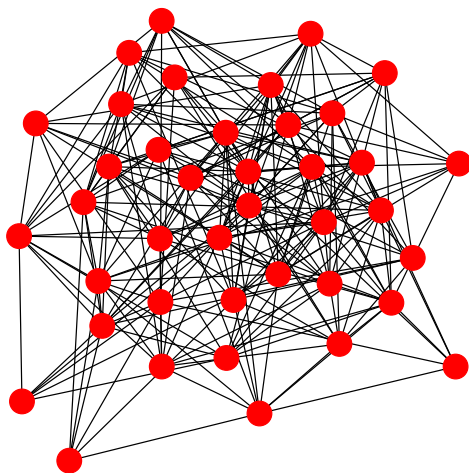


Figure 3: Random graph $G_{40,0.3}$ with 40 vertices and edge probability 0.3

4 Ramsey's Theorem

Random graphs can easily be scaled, and so it becomes increasingly difficult to form an explicit construction of a specific graph. Due to this, we become more interested in analysing the existence of specific properties than their explicit examples. In order to do this we use the probabilistic method.

The probabilistic method, as developed primarily by Paul Erdős, allows us to represent a problem in terms of an appropriate probability space. From this space we can pick a random object and calculate the probability that it satisfies the desired conditions. If it can be proved that the probability of the characteristic is strictly positive then the object must exist [3, Chapter 1].

4.1 Ramsey's Theorem

This method can be demonstrated through the famous Ramsey Theorem, assessing the relationship between set of independence numbers and the set of cliques in a random graph. Simplistically, it asserts that if a graph is large enough, both properties cannot be simultaneously small.

Theorem 4.1 (Ramsey's Theorem). *For every $k \geq 3$ there is some integer $n = n(k)$ such that if $v(G) \geq n$, then $\max\{\alpha(G), \omega(G)\} \geq k$.*

This theorem says that for every k , there exists an integer n such that every graph with n vertices must have either a clique or an independent set of size k .

4.1.1 Simulation

The following table represents a simulation that I created of randomly generated graphs of size n , investigating how often a graph with no clique or independent set of size $k = 4$ after 200 trials. It empirically shows that as the number of vertices increases, the probability that there is no clique or independent decreases until it reaches 0 (in this case at $n = 18$). Thus demonstrating that $k = 4$ satisfies Ramsey's Theorem.

Table 1: A table demonstrating Ramsey's Theorem for $k=4$ with varying n

n	#(has clique/indep)	Trials	P(no clique or indep)
5	1	200	0.995
6	1	200	0.995
7	5	200	0.975
8	14	200	0.93
9	29	200	0.855
10	54	200	0.73
11	92	200	0.54
12	118	200	0.41
13	145	200	0.275
14	167	200	0.165
15	183	200	0.085
16	191	200	0.045
17	198	200	0.01
18	200	200	0

The code I implemented to create this simulation is as follows:

I started by using the random graph generator defined in Listing 1, before starting an evaluation function that identified whether a clique or independent set of size k was found in a given random graph.

Listing 2: Clique and Independent Set Evaluator

```
def Eval_X(A, k):
    n=A.shape[0] #recalling n from above
    max_edges = comb(k,2) #no. edges in a k-clique
    for S in combinations(range(n), k):
        sub = A[np.ix_(S, S)]
        #creates subgraph similar to X but for subset S
        e = np.triu(sub,1).sum()
        #counts no. edges without double count
        if e == max_edges:
            return True #if S is a k-clique
        if e == 0:
            return True #if S is k-independent
    return False
```

I then looped this algorithm for varying values of n , keeping a fixed k , in order to estimate the probability that there was no clique or independent set of size k . This information could then be collected into a table as shown in 1.

The complete code is found below.

Listing 3: Ramsey Simulation

```
#Requires: Random Graph Generator (Listing 1)
```

Eval_X (Listing 2)

```

from tabulate import tabulate

def Ramsey_simulation(k, trials=200, seed=123):
    rng = np.random.default_rng(seed)
    #seed is an arbitrary value
    threshold_n = 2**(k/2 - 1)
    n_values = range(max(k, 5), min(k*4, 50))

    rows = []
    for n in n_values:
        bad = sum(
            Eval_X(X(n, rng=rng), k)
            for _ in range(trials))
        p_bad = bad / trials
        p_good = 1 - p_bad

        rows.append([n, bad, trials, f"{p_good:.3f}"])

        if p_good == 0:
            break
    # no point continuing once probability hits zero

    print(tabulate(rows,
                    headers=["n",
                             "#(has clique/indep)",
                             "Trials",
                             "P(no clique or indep)"],
                    tablefmt="latex"))

```

4.2 Ramsey Lower Bound

Now Ramsey's theorem has guaranteed that $n(k)$ exists, which means the set of integers n where every graph, of n vertices must contain a clique or independent set of size n , is non-empty. Since we know that it is not empty, it can also be said that it has a minimum. This minimum is defined

$$R(k) = \min\{n : \max\{\alpha(G), \omega(G)\} \geq k \text{ for every } n\text{-vertex graph } G\}$$

It is extremely difficult to try to determine the exact value of $R(k)$ for each k and so we are forced to rely on its bounds to examine the behaviour close to the threshold. During his early work on graph theory using the probabilistic method, Erdős proved the following exponential lower bound:

Theorem 4.2. *For every integer $k \geq 3$ we have $R(k) \geq 2^{k/2-1}$.*

Erdős has here stated that the smallest integer n such that every n -vertex graph has a clique or independent of size k is greater than $2^{\frac{k}{2}-1}$.

4.2.1 Simulation

This theorem is easy to observe in practice. As shown in the following table depicting a simulation I created to show the number of trials taken to find a clique or independent of size k in an n -vertex graph when n is the threshold given in the above theorem.

Table 2: Table showing whether a clique or independent set of size k was found at the threshold.

k	$n = \lceil 2^{\frac{k}{2}-1} \rceil$	Witness found	Trials
8	8	Yes	1
9	12	Yes	1
10	16	Yes	1
11	23	Yes	1

The fact that for each k for $k = 8, 9, 10, 11$, a witness is found after only a single trial corroborates the theorem.

This table has built on the previous 2 simulations mentioned:

First, I again used the Random Graph Generator found in Listing 1. Then this was combined with the Evaluation function found in Listing 2. This time I needed a function to determine the existence of a clique or independent set of size k at the boundary. The algorithm I created loops the Evaluation function many times for varying k , stopping when either the number trials had been reached or a witness was found. The number of vertices n in each graph as determined by the boundary found in Theorem 4.2. The completed programme follows:

Listing 4: Existence at Boundary Simulation

```
#Requires: Random Graph Generator (Listing 1)
          Eval_X (Listing 2)

from math import ceil
from tabulate import tabulate

def sim_existence_at_bound(k_values,
                           max_tries=100000,
                           seed=123):
    rows = []

    for k in k_values:
        rng = np.random.default_rng(seed + k)
```

```

#ensures different randomness for each k
n = ceil(2**(k/2 - 1))
#computes theoretical bound (rounds up)

tries = None
for t in range(1, max_tries + 1):
    if not Eval_X(X(n, rng=rng), k):
        #determines if k-clique/independent found
        tries = t
        break

found = tries is not None

rows.append([k,n,"Yes" if found else "No",
             tries if found else ">{max_tries}"])
print(tabulate(rows,
               headers=["k", "n = ceil(2^{k/2-1})",
                       "Witness found", "Trials"],
               tablefmt="latex"
            ))

```

4.2.2 Proof

Proof. Set $k \geq 3$ and let n be a positive integer. Now let

$$\Omega = \{\text{all graphs with vertex set } V = \{1, \dots, n\}\}.$$

Define a probability measure μ on Ω by

$$\mu(G) = \frac{1}{|\Omega|}.$$

Equivalently this is the graph $G(n, \frac{1}{2})$ by Proposition 3.1. We want to show that

$$\mathbb{P}(\alpha(G) < k \text{ and } \omega(G) < k) > 0.$$

For $X \subseteq V$ with $|X| = k$. The set X forms a clique if and only if all $\binom{k}{2}$ edges exist. Now since each edge forms independently with probability $1/2$, thus

$$P(X \text{ is a clique}) = 2^{-\binom{k}{2}}.$$

The statement " G has $\omega(G) \geq k$ " is equivalent to the statement "there exists a set $X \subseteq V$ with $|X| = k$ which is a clique". Hence,

$$\{\omega(G) \geq k\} = \bigcup_{\substack{X \subseteq V \\ |X|=k}} \{X \text{ is a clique}\},$$

so by the union bound,

$$\mathbb{P}(\omega(G) \geq k) \leq \sum_{|X|=k} \mathbb{P}(X \text{ is a clique}) = \binom{n}{k} 2^{-\binom{k}{2}}.$$

Now we can use the estimate $\binom{n}{k} \leq n^k$ to show,

$$\mathbb{P}(\omega(G) \geq k) \leq n^k 2^{-\frac{k(k-1)}{2}} = 2^{(\log_2 n)k - \frac{k(k-1)}{2}}.$$

Now if we set $n = 2^{\frac{k}{2}-1}$ then we can use the logarithm rule to say,

$$\log_2 n = \frac{k}{2} - 1,$$

which can be substituted into the power to give,

$$k\left(\frac{k}{2} - 1\right) - \frac{k(k-1)}{2} = \frac{k^2}{2} - k - \frac{k^2 - k}{2} = -\frac{k}{2}.$$

Therefore,

$$\mathbb{P}(\omega(G) \geq k) \leq 2^{-\frac{k}{2}}.$$

the set X forms an independent set if and only if none of the $\binom{k}{2}$ possible edges exist. Since each edge is absent independently with probability $1/2$,

$$\mathbb{P}(X \text{ is an independent set}) = 2^{-\binom{k}{2}}.$$

The statement “ G has $\alpha(G) \geq k$ ” is equivalent to the statement “there exists a set $X \subseteq V$ with $|X| = k$ which is an independent set”. Hence,

$$\{\alpha(G) \geq k\} = \bigcup_{\substack{X \subseteq V \\ |X|=k}} \{X \text{ is an independent set}\},$$

so by the union bound,

$$\mathbb{P}(\alpha(G) \geq k) \leq \sum_{|X|=k} \mathbb{P}(X \text{ is an independent set}) = \binom{n}{k} 2^{-\binom{k}{2}}.$$

Using the same estimates as before, $\binom{n}{k} \leq n^k$ and $n = 2^{\frac{k}{2}-1}$, we obtain

$$\mathbb{P}(\alpha(G) \geq k) \leq 2^{k\left(\frac{k}{2}-1\right) - \frac{k(k-1)}{2}} = 2^{-\frac{k}{2}},$$

We can combine these by again using the union bound,

$$\mathbb{P}(\alpha(G) \geq k \text{ or } \omega(G) \geq k) \leq 2 \cdot 2^{-\frac{k}{2}} = 2^{1-\frac{k}{2}}.$$

Since $k \geq 3$, we have $2^{1-\frac{k}{2}} < 1$. Thus,

$$\mathbb{P}(\alpha(G) < k \text{ and } \omega(G) < k) > 0.$$

Therefore, there exists at least one graph $G(2^{\frac{k}{2}-1}, \frac{1}{2})$ satisfying

$$\alpha(G) < k \quad \text{and} \quad \omega(G) < k.$$

□

5 Triangle Theorem

Theorem 5.1. $\hat{p} = n^{-1}$ is a threshold number for the existence of a triangle $G_{n,p}$.

Very basically, this theorem says that the probability that a given random graph contains a triangle changes drastically at the proposed threshold of $p = \frac{1}{n}$. It can be divided into 2 different cases: (1) when $p \ll \frac{1}{n}$ and (2) when $p \gg \frac{1}{n}$.

- if (1) is true then $P(G_{n,p} \text{ contains a triangle}) \rightarrow 0$.
- if (2) is true then $P(G_{n,p} \text{ contains a triangle}) \rightarrow 1$.

5.1 Triangle Theorem Simulation

In order to create more understanding of this theorem, I created an algorithm to show the probability of finding a triangle for a random graph in which p varies. This way we can see how the probability changes when it approaches the threshold. The table of results for $n = 50$ is shown below.

Table 3: A table to show the probability of finding a triangle in a graph $G_{50,p}$

p	Triangles found	Trials	P(triangle)	
0.005	0	200	0	
0.01	3	200	0.015	
0.015	13	200	0.065	
0.02	32	200	0.16	$\leftarrow 1/n$
0.025	44	200	0.22	
0.03	74	200	0.37	
0.04	124	200	0.62	
0.05	187	200	0.935	
0.075	200	200	1	
0.1	200	200	1	
0.15	200	200	1	
0.2	200	200	1	

Here we can see that the threshold number is highlighted at $p = 0.02$ where the probability is 0.16. Both above and below $p = 0.02$ the probability moves away from 0.16 at a sharp rate before tapering off as it approaches a probability of 1 and 0.

The code I used for this simulation again began with the random graph function shown in Listing 1. Then I created a function that determines whether the graph contains at least 1 triangle. I optimised it using graph theory, using the knowledge that if A is the adjacency matrix of a graph then entry (i, j) of A^k counts the number of walks of length k from vertex i to vertex j . Thus, the diagonal entries of A^3 count the number of walks of length 3 that start and end at the same point. Since we have 3 starting vertex in a triangle and 2 directions

of travel, our function essentially calculates $6x$ the number of triangles, which is effective as we are only returning true or false.

Listing 5: Triangle Detection

```
def has_triangle(A):
    # trace of A^3 counts closed walks of length 3
    # each triangle is counted 6 times (3 vertices, 2 directions)
    return np.trace(np.linalg.matrix_power(A, 3)) > 0
#replaced with the trace of A^3 for optimisation purposes,
#as this equals 6 x the no. of triangles

def table_triangles(n=50, trials=200, seed=123):
    rng = np.random.default_rng(seed)
    threshold = 1/n
    p_values = [0.005, 0.01, 0.015, round(threshold, 3),
                0.025, 0.03, 0.04, 0.05, 0.075,
                0.1, 0.15, 0.2]
```

In order to show the activity at the threshold I looped this function over varying values of p , maintaining the same n . Whilst highlighting the threshold to show how the probability varies as p moves away from it.

Listing 6: Triangle Threshold Simulation

```
#Requires: Random Graph Generator (Listing 1)
          Triangle Detection (Listing 5)

rows = []
for p in p_values:
    successes = sum(
        has_triangle(X(n, p=p, rng=rng))
        for _ in range(trials)
    )
    p_hat = successes / trials
    at_threshold = "<- 1/n" if p == round(threshold, 3) else ""
    rows.append([f"{p:.3f}", successes, trials,
                f"{p_hat:.3f}", at_threshold])

print(f"\nTriangle threshold in G(n={n},p),
      theoretical threshold p=1/{n}={threshold:.3f}")
print(tabulate(rows,
               headers=["p", "Triangles found",
                       "Trials", "P(triangle)", ""],
               tablefmt="latex_booktabs"))

table_triangles(n=50, trials=200, seed=123)
```

5.2 Triangle Theorem Proof

Proof. Let $G = G_{n,p}$ be the random graph on n vertices with an independent edge probability $p \in [0, 1]$. Let $X = \sum_{i=1}^{\binom{n}{3}} X_i$ where i is a triple of vertices in the random graph and X_i is the indicator of whether the triple i forms a triangle, with $X_i = 1$ if it is a triangle and $X_i = 0$ if not. Thus, X counts the total number of triangles in $G_{n,p}$.

By the linearity of expectation we have that:

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^{\binom{n}{3}} X_i\right] = \sum_{i=1}^{\binom{n}{3}} \mathbb{E}[X_i] = \binom{n}{3} \cdot p^3.$$

By the definition of the binomial coefficient,

$$\binom{n}{3} = \frac{n(n-1)(n-2)}{6} = \frac{n^3 - 3n^2 + 2n}{6} = \frac{n^3}{6} + O(n^2)$$

Thus, we can say

$$\mathbb{E}[X] = \frac{n^3 p^3}{6} + O(n^2 p^3).$$

Now for (1), let $p \ll \frac{1}{n}$. It is clear that the probability that there is a triangle is the same as $\mathbb{P}(X \geq 1)$, so using Markov's inequality we have

$$\mathbb{P}(X \geq 1) \leq \mathbb{E}[X].$$

Since $p \ll \frac{1}{n}$, both $n^3 p^3 \rightarrow 0$ and $O(n^2 p^3) \rightarrow 0$, so $\mathbb{E}[X] \rightarrow 0$. Thus,

$$\mathbb{P}(X \geq 1) \rightarrow 0$$

when $p \ll \frac{1}{n}$.

Now for (2), suppose $p \gg \frac{1}{n}$, then $n^3 p^3 \rightarrow \infty$. Hence, $\mathbb{E}[X] \rightarrow \infty$. The fact that the expectation is large doesn't prove the probability that a triangle exists is large. We must show that $\mathbb{P}(X > 0) \rightarrow 1$. By the second moment method,

$$\mathbb{P}(X > 0) \geq \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}.$$

To compute $\mathbb{E}[X^2]$, expand

$$\mathbb{E}[X^2] = \mathbb{E}\left[\left(\sum_i X_i\right)\left(\sum_j X_j\right)\right] = \sum_{i,j} \mathbb{E}[X_i X_j].$$

Hence, this will expand to incorporate the 4 cases:

$$\mathbb{E}[X^2] = \sum_{\substack{i,j \\ s=0}} \mathbb{E}[X_i X_j] + \sum_{\substack{i,j \\ s=1}} \mathbb{E}[X_i X_j] + \sum_{\substack{i,j \\ s=2}} \mathbb{E}[X_i X_j] + \sum_{\substack{i,j \\ s=3}} \mathbb{E}[X_i X_j],$$

where $s = i \cap j$ describing the number of shared vertices between i and j . First, when i and j share a maximum of 1 vertex they share no edges not already counted. This makes X_i and X_j independent, which gives

$$\mathbb{E}[X_i X_j] = \mathbb{E}[X_i] \mathbb{E}[X_j] = p^6,$$

Using the observation that $\binom{n}{k} = \frac{n^k}{k!} + O(n^{k-1})$, we can calculate the contributions for $s = 0$ and $s = 1$. When $s = 0$ we have

$$\binom{n}{3} \binom{n-3}{3} = \frac{n^6}{36} + O(n^5),$$

when $s = 1$ we have:

$$n \cdot \binom{n-1}{2} \binom{n-3}{2} = \frac{n^5}{4} + O(n^4).$$

Therefore, the overall contribution for $s = 0$ and $s = 1$ is

$$p^6 \left(\frac{n^6}{36} + O(n^5) \right) + \frac{n^5}{4} + O(n^4) = \frac{n^6 p^6}{36} + O(n^5 p^6).$$

The next case is when $s = 2$, meaning there are 4 vertices and 5 possible edges. Thus,

$$\begin{aligned} \sum_{\substack{i,j \\ s=2}} \mathbb{E}[X_i X_j] &= p^5 \left(\frac{\binom{n}{2} (n-2)(n-3)}{2} \right) = p^5 \left(\frac{(\frac{n^2}{2} + O(n))(n-2)(n-3)}{2} \right) \\ &= p^5 \left(\frac{\frac{n^4}{2} - \frac{5n^3}{2} + 3n^2 + O(n^3)}{2} \right) = \frac{n^4 p^5}{4} + O(n^3 p^5). \end{aligned}$$

The final case of $s = 3$ is when i and j are identical, making

$$\mathbb{E}[X_i X_j] = \mathbb{E}[X^2] = \mathbb{E}[X] = p^3,$$

and this occurs in $\binom{n}{3}$ different ways, thus case 3 has a contribution of

$$\frac{n^3 p^3}{6} + O(n^2 p^3).$$

When all the cases are combined, the result is

$$\begin{aligned} \mathbb{E}[X^2] &= \frac{n^6 p^6}{36} + O(n^5 p^6) + \frac{n^4 p^5}{4} + O(n^3 p^5) + \frac{n^3 p^3}{6} + O(n^2 p^3) \\ &= \frac{n^6 p^6}{36} + O(n^5 p^6) + O(n^4 p^5) + O(n^3 p^3) \end{aligned}$$

Now it is clear that

$$\mathbb{E}[X]^2 = \left(\frac{n^3 p^3}{6} + O(n^2 p^3) \right)^2 = \left(\frac{n^3 p^3}{6} \right)^2 + 2 \cdot \left(\frac{n^3 p^3}{6} \right) \cdot O(n^2 p^3) + (O(n^2 p^3))^2 = \frac{n^6 p^6}{36} + O(n^5 p^6).$$

So,

$$\begin{aligned}\frac{\mathbb{E}[X^2]}{\mathbb{E}[X]^2} &= \frac{\frac{n^6 p^6}{36} + O(n^5 p^6) + O(n^4 p^5) + O(n^3 p^3)}{\frac{n^6 p^6}{36} + O(n^5 p^6)} \\ &= 1 + O\left(\frac{1}{n}\right) + O\left(\frac{1}{np}\right) + O\left(\frac{1}{n^3 p^3}\right).\end{aligned}$$

Since $p \gg \frac{1}{n}$, all error terms tend to 0. Therefore,

$$\mathbb{P}(X > 0) \geq \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]} \rightarrow 1.$$

□

Appendix

A Clique Visualisation

```
import networkx as nx
import matplotlib.pyplot as plt
import numpy as np

def X(n, p):
    upper = np.random.binomial(1, p, size=(n, n))
    A = np.triu(upper, 1)
    A = A + A.T #the matrix is now symmetric
    return A

G = nx.from_numpy_array(X(3,1))
plt.figure(figsize=(3,3))
nx.draw(
    G,
    node_color="red",
    edge_color="black",
    node_size=600
)

plt.show()
plt.close()
```

B Independent Set Visualisation

```
import networkx as nx
import matplotlib.pyplot as plt
import numpy as np

def X(n, p):
    upper = np.random.binomial(1, p, size=(n, n))
    A = np.triu(upper, 1)
    A = A + A.T
    return A

G = nx.from_numpy_array(X(3,0.0))
plt.figure(figsize=(4,4))
nx.draw(
    G,
    node_color="red",
```

```

        edge_color="black",
        node_size=600
    )
    plt.show()
    plt.close()

```

C Random Graph Visualisation

```

import numpy as np
import networkx as nx
import matplotlib.pyplot as plt

def X(n, p):
    upper = np.random.binomial(1, p, size=(n, n))
    A = np.triu(upper, 1)
    A = A + A.T
    return A

G = nx.from_numpy_array(X(40, 0.3))
plt.figure(figsize=(6, 6))
nx.draw(
    G,
    node_color="red",
    edge_color="black",
    node_size=400
)
plt.show()
plt.close()

```

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